Royaume du Maroc

Ministère de l'Aménagement du Territoire National,
de l'Urbanisme, de l'Habitat
et de la Politique de la Ville



المملكة المغربية

وزارة إعداد التراب الوطني والتعمير والإسكان وسياسة المدينة

# RAPPORT D'ANALYSE

# Comparative Analysis Report

Rapport généré automatiquement

| | |
|---|---|
| **Organisme** | Ministère de l'Habitat |
| **Date de génération** | 10/08/2026 à 02:06 |
| **Plateforme** | Immobilier IA |

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# INFORMATIONS DU RAPPORT

Cette page présente les principales informations relatives au rapport généré automatiquement par Immobilier IA.

| Champ | Valeur |
|---|---|
| Titre du rapport | Comparative Analysis Report |
| Sous-titre | Rapport généré automatiquement |
| Auteur | Ministère de l'Habitat |
| Date de génération | 10/08/2026 02:06 |
| Nombre de sections | 4 |
| Nombre de tableaux | 1 |
| Nombre de graphiques | 1 |
| Lignes SQL | 15 |
| Colonnes SQL | 4 |
| Source des données | SQL Server - Ministère de l'Habitat |
| Plateforme | Immobilier IA |

Les informations ci-dessus sont générées automatiquement au moment de la création du rapport afin d'assurer la traçabilité et la compréhension du document.

# 1. Objet de la demande

Fais un rapport sur les prix des matériaux

## Indicateurs clés

| Résultats | Minimum | Maximum |
|---|---|---|
| 10000 lignes | 1,44 | 11 249,59 |
| Nombre de lignes retournées par la requête | Valeur minimale de PrixMoyenMAD | Valeur maximale de PrixMoyenMAD |

| Moyenne | Médiane |
|---|---|
| 455,61 | 39,82 |
| Moyenne de PrixMoyenMAD | Médiane de PrixMoyenMAD |

# 2. Résumé et analyse

# Executive Summary

This report analyzes the prices of materials (PrixMoyenMAD) across Regions and Provinces, categorized by material type. Based on 10,000 observations, the dataset shows substantial price variation with a high standard deviation relative to the average. The mean price is 455.61 MAD, while the median is 39.82 MAD, indicating a right-skewed distribution with notable outliers. There is no supported inference of causal trends due to the provided data. Visualizations are recommended to illustrate geographic distribution and regional price levels.

# Objective

• Assess the price distribution of construction/materials (PrixMoyenMAD) by Region, Province, and Material type.

• Identify central tendency and dispersion characteristics.

• Provide visualization and recommendations to support market intelligence and planning decisions.

# Dataset Overview

• Structure: 10,000 rows with 4 columns.

• Columns:

• Region (categorical)

• Province (categorical)

• Materiau (categorical)

• PrixMoyenMAD (numeric)

• Geographic dimensions are present via Region and Province; material dimension via Materiau.

• The dataset provides a broad cross-section of prices, allowing geographic and material-level comparisons but does not include time-series or volume data.

# Methodology

• The report is generated from SQL query results and statistical analysis of PrixMoyenMAD.

• Descriptive statistics computed for PrixMoyenMAD:

• minimum, maximum, average, median, standard deviation, and total sum.

• Geographic and categorical analyses are used to inform recommended visualizations.

# Statistical Analysis

• PrixMoyenMAD (MAD)

• Minimum: 1.44

• Maximum: 12,499.59

• Average (mean): 455.61

• Median: 39.82

• Standard deviation: 1,596.32

• Total: 2,129,983.21

• Interpretation:

• The mean price (455.61 MAD) is substantially higher than the median (39.82 MAD), indicating a positively skewed distribution with potential high-price outliers.

• The very large standard deviation (1,596.32) relative to the mean suggests wide price dispersion and significant variability across observations.

• The wide range (1.44 to 12,499.59) reinforces the presence of extreme values that influence the average.

• Anomalies:

• The coexistence of a relatively low median and a very high maximum implies outliers or a subset of very expensive material prices. This pattern aligns with the high standard deviation and the large gap between mean and median.

# Key Findings

• Price dispersion is substantial: PrixMoyenMAD varies widely across records.

• Central tendency indicates skewness: median is far below mean, suggesting a minority of very high-priced entries heavily influence the average.

• Geographic data structure exists (Region and Province), enabling geographic visualization of price levels.

• Material variations are present but not quantified in the provided statistics beyond the aggregated PrixMoyenMAD.

# Recommended Visualizations

• Map for PrixMoyenMAD by Region

• Rationale: Geographic columns detected; spatial visualization should reveal regional price patterns and identify potential regional disparities.

• Bar chart by Region

• Rationale: Categorical (Region) vs. numeric (PrixMoyenMAD) to compare average price levels across regions and highlight regional disparities.

# Business Recommendations

• Investigate regional price dispersion:

• Regions with notably higher average PrixMoyenMAD should be examined for market structure, material availability, or logistical factors contributing to higher prices.

• Regions with lower prices may indicate competitive markets or material abundance; investigate potential for price normalization or competition-based pricing strategies.

• Address outliers in pricing:

• The presence of high-price outliers may distort overall pricing strategies. Consider segmentation by Material and Region to understand whether extreme values are material-specific or region-specific.

• Material-level analysis (beyond aggregated PrixMoyenMAD):

• Break down PrixMoyenMAD by Materiau within Regions to identify which materials drive price volatility and where supply constraints may exist.

• Data quality and monitoring:

• Given the high dispersion, ongoing collection of time-series data could help assess price evolution, detect trend patterns, and support forecasting for planning and procurement.

# Conclusion

The analysis reveals substantial variability in material prices across Regions and Provinces, with a pronounced right-skew indicated by the gap between mean and median and a high standard deviation. While the data provide a solid basis for geographic visualization and regional price comparison, no time-based trends or causative factors are inferable from the current dataset. The recommended visualizations will aid in identifying geographic disparities, and the business recommendations focus on understanding and mitigating price dispersion, informed by region- and

material-level insights.

# 3. Requête SQL exécutée

SELECT r.nom_region AS Region, p.nom_province AS Province, m.nom AS Materiau, AVG(f.prix_moyen) AS PrixMoyenMAD FROM FACT_PRIX_INDICES f INNER JOIN REGION r ON f.id_region = r.id_region INNER JOIN PROVINCE_PREFECTURE p ON f.id_province = p.id_province INNER JOIN NIVEAU_MATERIAU m ON f.id_niveau_mat = m.id GROUP BY r.nom_region, p.nom_province, m.nom ORDER BY Materiau, Region, Province

# 4. Méthodologie

Ce rapport est généré automatiquement à partir des données disponibles dans la base de données du Ministère. La question formulée en langage naturel est interprétée par l'agent d'intelligence artificielle, puis transformée en requête SQL. Les résultats retournés sont ensuite analysés afin de produire une synthèse, des indicateurs clés et des visualisations.

# 5. Conclusion

Les résultats présentés dans ce rapport permettent de synthétiser les données retournées par la requête et d'en faciliter l'interprétation à travers des indicateurs et des visualisations adaptés.

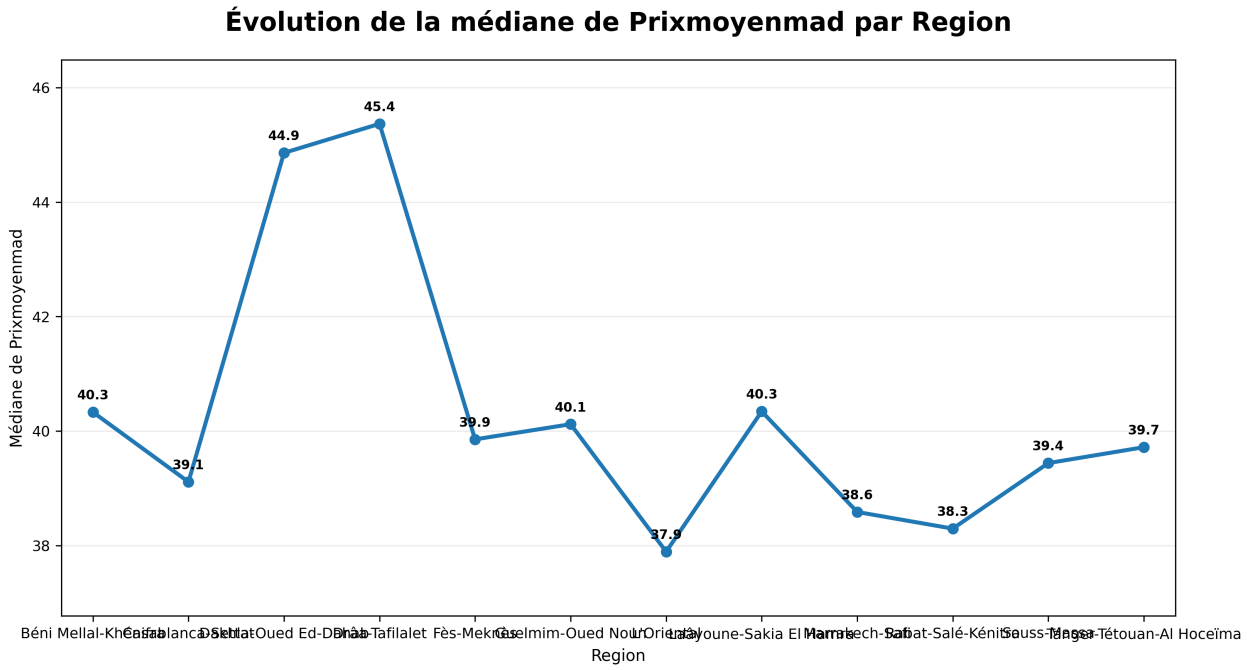
# 6. Aperçu des résultats (15 lignes sur 10000)

| Region | Province | Materiau | PrixMoyenMAD |
|---|---|---|---|
| Béni Mellal-Khénifra | Azilal | 1 Compartiment (u) | 393,88 |
| Béni Mellal-Khénifra | Béni Mellal | 1 Compartiment (u) | 380,02 |
| Béni Mellal-Khénifra | Fquih Ben Salah | 1 Compartiment (u) | 377,55 |
| Béni Mellal-Khénifra | Khénifra | 1 Compartiment (u) | 382,58 |

| Region | Province | Materiau | PrixMoyenMAD |
|---|---|---|---|
| Béni Mellal-Khénifra | Khouribga | 1 Compartiment (u) | 323,67 |
| Casablanca-Settat | Benslimane | 1 Compartiment (u) | 253,70 |
| Casablanca-Settat | Berrechid | 1 Compartiment (u) | 254,80 |
| Casablanca-Settat | Casablanca | 1 Compartiment (u) | 256,67 |
| Casablanca-Settat | El Jadida | 1 Compartiment (u) | 253,55 |
| Casablanca-Settat | Médiouna | 1 Compartiment (u) | 260,32 |
| Casablanca-Settat | Mohammadia | 1 Compartiment (u) | 265,20 |
| Casablanca-Settat | Nouaceur | 1 Compartiment (u) | 264,94 |
| Casablanca-Settat | Settat | 1 Compartiment (u) | 282,87 |
| Casablanca-Settat | Sidi Bennour | 1 Compartiment (u) | 267,23 |
| Dakhla-Oued Ed-Dahab | Aousserd | 1 Compartiment (u) | 315,33 |

# 7. Visualisations

Les visualisations suivantes présentent les tendances et comparaisons principales identifiées à partir des résultats.




*Évolution annuelle de la médiane de Prixmoyenmad sur la période étudiée.*